

AWS S3 + CloudFront Static Website Hosting

Project Overview

This project demonstrates how to host a static website using Amazon S3 and deliver content securely through Amazon CloudFront.

The website files (HTML, CSS, and JavaScript) are stored in an S3 bucket, while CloudFront is used as a Content Delivery Network (CDN) to provide faster content delivery, HTTPS support, and secure access to the website.

Architecture

```
graph TD; User --> CloudFront; CloudFront --> AmazonS3Bucket[Amazon S3 Bucket];
```

User
↓
CloudFront
↓
Amazon S3 Bucket

AWS Services Used

- Amazon S3
 - Amazon CloudFront
 - Origin Access Control (OAC)
 - AWS IAM
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Features

- Static website hosting using Amazon S3
 - Global content delivery using CloudFront
 - Secure access with Origin Access Control (OAC)
 - HTTPS support through CloudFront
 - Improved website performance using CDN caching
 - Private S3 bucket access through CloudFront
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Project Steps

1. Created an Amazon S3 Bucket

- Created an S3 bucket to store website files.
- Uploaded HTML, CSS, and JavaScript files.

2. Enabled Static Website Hosting

- Configured S3 bucket for static website hosting.
- Defined the index document as `index.html`.

3. Created a CloudFront Distribution

- Created a CloudFront distribution.
- Connected the S3 bucket as the origin.

4. Configured Origin Access Control (OAC)

- Created and attached an Origin Access Control.
- Allowed CloudFront to securely access the S3 bucket.

5. Updated Bucket Policy

- Added permissions allowing CloudFront to access S3 objects.
- Restricted direct public access to bucket content.

6. Tested Website Access

- Verified website accessibility through the CloudFront URL.
- Confirmed content delivery through CloudFront.

Project Structure

```
aws-s3-cloudfront-static-website/  
|  
├─ index.html  
├─ style.css  
├─ script.js  
├─ screenshots/  
|   └─ s3-bucket.png  
|   └─ cloudfront-distribution.png  
|   └─ website-output.png  
└─ README.md
```

Skills Learned

- Amazon S3 Bucket Management
- Static Website Hosting
- CloudFront CDN Configuration
- Origin Access Control (OAC)
- Bucket Policy Management
- Content Delivery Networks (CDN)
- AWS Security Best Practices

Challenges Faced

- Troubleshooting S3 object access permissions.
- Configuring bucket policies for CloudFront access.
- Resolving CloudFront "Access Denied" errors.
- Understanding OAC and secure S3 access.

Outcome

Successfully hosted a static website using Amazon S3 and Amazon CloudFront with secure access configuration and global content delivery.

Author

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